

Chemistry — Assignment 8: Hybridisation & Bond Fission

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Chemistry (Foundational Chemistry) · Semester 5

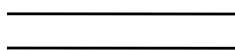
Question. Explain the types of hybridisation with examples. Differentiate homolytic vs heterolytic fission.

Hybridisation = mixing of atomic orbitals of nearly equal energy into equivalent hybrids.

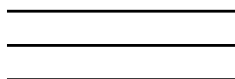
- **sp³:** $1s+3p \rightarrow 4$ hybrids, tetrahedral, 109.5° , 4 σ (methane, ethane)
- **sp²:** $1s+2p \rightarrow 3$ hybrids, trigonal planar, 120° , 3 σ + 1 π (ethene, benzene)
- **sp:** $1s+1p \rightarrow 2$ hybrids, linear, 180° , 2 σ + 2 π (ethyne, CO₂)



methane (sp³)



ethene (sp²)



ethyne (sp)

Homolytic fission	Heterolytic fission
bond breaks evenly; each atom keeps one electron → free radicals	bond breaks unevenly; one atom takes both electrons → ions (cation + anion)
fishhook (single-barb) arrows; favoured by heat/UV, nonpolar	full curved arrow; favoured by polar solvents